

$$\begin{array}{llll}
 \text{minimizar} & 2x_1 & +3x_2 & +4x_3 \\
 \text{s.a} & x_1 & +2x_2 & +x_3 \geq 3 \\
 \text{Primal} & 2x_1 & -x_2 & +3x_3 \geq 4 \\
 & & & x_i \geq 0
 \end{array}$$

Tabla 3: Tabla de Tucker de dualidad

minimización		maximización	
restricciones	$\geq$	≥ 0	variables
	$\leq$	≤ 0	
	$=$	irrestringido	
variables	≥ 0	$\leq$	restricciones
	≤ 0	$\geq$	
	irrestringido	$=$	

$$\text{Max } 3\gamma_1 + 4\gamma_2 \quad (\text{DUAL})$$

$$\gamma_1 + 2\gamma_2 \leq 2$$

$$2\gamma_1 - \gamma_2 \leq 3$$

$$\gamma_1 + 3\gamma_2 \leq 4$$

$$-3\gamma_1 - 4\gamma_2$$

$$\gamma_1 + 2\gamma_2 + s_1 = 2$$

$$2\gamma_1 - \gamma_2 + s_2 = 3$$

$$\gamma_1 + 3\gamma_2 + s_3 = 4$$

$$\gamma_1, \gamma_2 \geq 0$$

$$\gamma_1, \gamma_2, s_i \geq 0$$

$$B = (s_1, s_2, s_3)$$

	γ_1	γ_2	s_1	s_2	s_3	
	-3	-4	0	0	0	0 + 2b ₁
s_1	1	2	1	0	0	2 :2
s_2	2	-1	0	1	0	3 + 1/2 f ₁
s_3	1	3	0	0	1	4 - 3/2 f ₁

	γ_1	γ_2	s_1	s_2	s_3	
	-1	0	2	0	0	4 + 2/5 f ₃
γ_2	1/2	1	1/2	0	0	1 - 1/5 f ₃
s_2	5/2	0	1/2	1	0	7 - 3/5 f ₃
s_3	-1/2	0	9/2	0	1	1 + 1/5 f ₃

	γ_1	γ_2	s_1	s_2	s_3	
	0	0	11/5	2/5	0	28/5 → -Z
γ_2	0	1	1/10	-1/5	0	1/4
γ_1	1	0	1/5	2/5	0	8/5
s_3	0	0	46/10	1/5	1	9/5

good

$$\text{Min } 160X_1 + 120X_2 + 280X_3$$

$$\text{s.a. } 2X_1 + X_2 + 4X_3 \geq 1$$

$$2X_1 + 2X_2 + 2X_3 \geq 3/2$$

$$X_1 \geq 0, X_2 \geq 0, X_3 \geq 0$$

$$(2X_1 + X_2 + 4X_3 - 1) \cdot \gamma_1 = 0$$

$$(2X_1 + 2X_2 + 2X_3 - 3/2) \cdot \gamma_2 = 0$$

$$(\gamma_1 + \gamma_2 - 80)X_1 = 0 \rightarrow X_1 \geq 0$$

$$(\gamma_1 + 2\gamma_2 - 120)X_2 = 0 \rightarrow X_2 \geq 0$$

$$(2\gamma_1 + \gamma_2 - 140)X_3 = 0$$

$$\downarrow$$

$$X_3 = 0$$

$$(2X_1 + X_2 - 1) \cdot \cancel{X_0} = 0$$

$$(2X_1 + 2X_2 - 3/2) \cdot \cancel{X_0} = 0$$

$$2X_1 + X_2 = 1$$

$$2X_1 + 2X_2 = 3/2$$

$$X_2 = 1 - 2X_1$$

$$2X_1 + 2 - 4X_1 = 3/2$$

$$2 - 3/2 = 2X_1$$

$$X_1 = 1/4$$

$$\downarrow$$

$$\therefore X_2 = 1/2$$

$$\text{Max } \gamma_1 + \frac{3}{2}\gamma_2$$

$$\rightarrow \gamma_1 + \gamma_2 \leq 80$$

$$\gamma_1 + 2\gamma_2 \leq 120$$

$$2\gamma_1 + \gamma_2 \leq 140$$

$$\gamma_i \geq 0$$

$\downarrow$

$$\text{Min } -\gamma_1 - \frac{3}{2}\gamma_2$$

$$\gamma_1 + \gamma_2 + s_1 = 80$$

$$\gamma_1 + 2\gamma_2 + s_2 = 120$$

$$2\gamma_1 + \gamma_2 + s_3 = 140$$

	-1	-3/2	0	0	0	0	+ 3/4 · F ₃
s ₁	2	2	1	0	0	160	-F ₃
s ₂	1	2	0	1	0	120	:2
s ₃	4	2	0	0	1	280	-F ₃

	-1/4	0	0	3/4	0	90	+ 1/4 F ₂
s ₁	1	0	1	-1	0	40	
γ ₂	1/2	1	0	1/2	0	60	-1/2 F ₂
s ₃	3	0	0	-1	1	160	-3 F ₂

	0	0	1/4	1/2	0	100
γ ₁	1	0	1	-1	0	40
γ ₂	0	1	-1/2	1	0	40
s ₃	0	0	-3	2	1	40